THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS X — SCIENCE (086) · FULL SYLLABUS PRACTICE PAPER

Session 2026–27 | CBSE / RBSE Pattern | Section A Biology · Section B Chemistry · Section C Physics

Class: 10	Subject: Science	Set 04	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.
2. All sections are compulsory. However, internal choice is provided in some questions. A student is expected to attempt only one of these questions.
3. Draw neat, labelled diagrams wherever required. Use of calculator is not allowed.

SECTION A — BIOLOGY — 16 questions · 30 marks

Q1–Q7 MCQs. Q8–Q9: Assertion (A) / Reason (R) – choose A, B, C or D.

1. Haemoglobin in red blood cells has a very high affinity for oxygen. In which organ do the red blood cells pick up the maximum amount of oxygen? **1**

A. Kidney B. Lungs C. Liver D. Heart

2. Which of the following statements about guard cells is correct? **1**

A. Water entering guard cells makes them swell and the stomatal pore opens

B. Water leaving guard cells makes the stomatal pore open

C. Guard cells are present only in roots

D. Guard cells have no chloroplasts

3. Taste and smell are detected by gustatory and olfactory receptors. The impulses from these receptors are interpreted in the: **1**

A. spinal cord B. cerebellum

C. forebrain (cerebrum) D. medulla

4. After a meal the blood sugar level of a healthy person rises. In response, the body releases the hormone: **1**

A. glucagon from the liver B. insulin from the pancreas

C. adrenaline from the adrenal gland D. thyroxine from the thyroid

5. In a mango flower, after fertilisation the part that develops into the seed is the: **1**

A. ovary B. ovule C. style D. stigma

6. Round yellow pea plants (RRYY) are crossed with wrinkled green plants (rryy) and the F_1 is self-pollinated. Out of 160 F_2 plants, the expected number of wrinkled yellow plants is: **1**

A. 10 B. 30 C. 90 D. 50

7. Which of the following represents a correct food chain of a pond ecosystem? **1**

A. Algae → water flea → fish → heron B. Heron → fish → water flea → algae

C. Fish → algae → heron → water flea D. Water flea → heron → algae → fish

8. **Assertion (A):** Blood flows in only one direction through the human heart. **1**
Reason (R): Valves in the heart prevent the backward flow of blood.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

9. **Assertion (A):** Thyroxine regulates the metabolism of carbohydrates, proteins and fats. **1**
Reason (R): Iodine is necessary for the synthesis of thyroxine.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

10. Distinguish between aerobic and anaerobic respiration on the basis of (i) the amount of energy released and (ii) the end products. **2**

11. Name the hormones responsible for (i) controlling blood sugar and (ii) the changes seen in boys at puberty. State the glands that secrete them. **2**

OR

What is a feedback mechanism in hormone regulation? Explain with one example.

12. State the 10% law. If the producers in an ecosystem receive 5000 J of energy, how much energy will be available to the secondary consumers? **2**

13. What are contraceptives? Explain any two different types of contraceptive methods and state one advantage of each. **3**

14. How is the sex of a child determined in human beings? Explain with the help of a flow chart. **3**

15. In a school garden, students observe that the leaves of a 'touch-me-not' plant droop when touched, that the tendrils of a pea plant coil around a stick, and that the roots of a germinating seed grow downwards while the shoot grows upwards.

(a) Name the type of movement shown by the tendrils of the pea plant. 1

(b) Which of the movements observed is not dependent on growth? 1

(c) Explain how a plant hormone causes the tendril to coil. 2

OR

(d) Name the tropic movement shown by roots and shoots and the hormone involved in the phototropic response.

16. (i) Draw a labelled diagram of the human heart and trace the path of blood from the right atrium back to the right atrium. (ii) Why is the human heart four-chambered? **5**

OR

(i) What is an ecosystem? Name its two components. (ii) How is ozone formed in the upper atmosphere? (iii) How do CFCs damage the ozone layer and why is this a concern?

SECTION B — CHEMISTRY — 12 questions · 25 marks

Q17–Q21 MCQs. Q22: Assertion (A) / Reason (R) – choose A, B, C or D.

17. Fat and oil containing foods become rancid on keeping for long. Which of the following can be added to prevent this? **1**

A. a catalyst B. an antioxidant C. an acid D. a base

18. The balanced form of the chemical equation $Al + O_2 \rightarrow Al_2O_3$ is: **1**

A. $2Al + 3O_2 \rightarrow Al_2O_3$ B. $4Al + 3O_2 \rightarrow 2Al_2O_3$

C. $Al + O_2 \rightarrow 2Al_2O_3$ D. $2Al + O_2 \rightarrow Al_2O_3$

19. A solution turns red litmus blue and has a pH of nearly: **1**

A. 2 B. 5 C. 7 D. 10

20. A student drops pieces of Mg, Zn, Fe and Cu in dilute HCl taken in four different test tubes. In which test tube would no bubbles be seen? **1**

A. Mg B. Zn C. Fe D. Cu

21. The functional group present in ethanal (CH_3CHO) is: **1**

A. alcohol (–OH) B. aldehyde (–CHO)

C. ketone (>C=O) D. carboxylic acid (–COOH)

22. **Assertion (A):** The blue colour of copper sulphate solution fades when an iron nail is dipped in it. **1**
Reason (R): Iron is more reactive than copper and displaces it from its salt solution.

A. Both A and R are true, and R is the correct explanation of A.

B. Both A and R are true, but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

23. What is observed when aqueous solutions of lead nitrate and potassium iodide are mixed? Write the balanced chemical equation and name the type of reaction. **2**

24. What happens when gypsum is heated at 373 K? Why should Plaster of Paris be stored in a moisture-proof container? **2**

25. What is an alloy? Name the alloys made of (i) copper and zinc, (ii) lead and tin. State one reason for making alloys. **3**

OR

Write balanced chemical equations for the reactions of (i) iron with steam, (ii) calcium with cold water and (iii) magnesium with dilute HCl.

26. Write the name and formula of the main product formed when: (a) ethanol is heated with excess concentrated H_2SO_4 at 443 K, (b) ethanoic acid reacts with sodium hydroxide, (c) ethene reacts with hydrogen in the presence of nickel. **3**

27. Anil often feels a burning sensation in the stomach after a heavy meal. The doctor explains that the stomach produces hydrochloric acid of pH about 1.5 to 3.0, which helps in digestion, but excess acid causes acidity. He suggests a tablet containing magnesium hydroxide (milk of magnesia) as an antacid.

(a) In what pH range is the acid present in the stomach? Is it strongly acidic or weakly acidic? 1

(b) Name the type of reaction between the antacid and the stomach acid. 1

(c) Write the balanced chemical equation for the reaction between magnesium hydroxide and hydrochloric acid. 2

OR

(d) Why should an antacid not be a strong base such as sodium hydroxide? Name one other common antacid.

28. Describe the extraction of copper from copper glance (Cu_2S): write the chemical equations for roasting and for the reduction step, and describe the electrolytic refining of copper (anode, cathode and electrolyte). **5**

OR

(a) Draw the structures of the three isomers of pentane (C_5H_{12}) and name them. (b) Write the balanced equation for the combustion of methane. (c) What is a substitution reaction? Give the equation for the reaction of methane with chlorine in sunlight (first step).

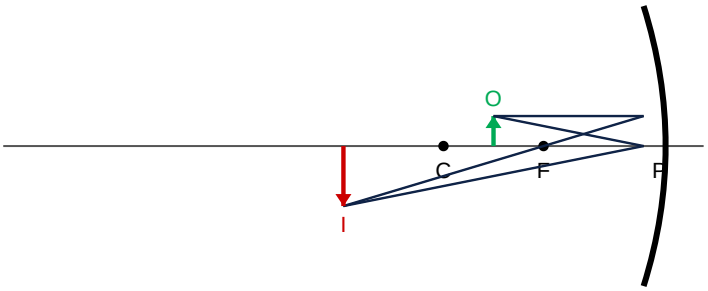
SECTION C — PHYSICS — 11 questions · 25 marks

Q29–Q32 MCQs. Q33: Assertion (A) / Reason (R) – choose A, B, C or D.

29. A ray of light travelling in air is incident on the surface of a glass slab. At which angle of incidence will the ray pass into glass without any deviation? **1**

A. 0° B. 30° C. 45° D. 90°

30. A concave mirror of focal length 10 cm forms an image of an object placed 15 cm from it, as shown. The image is: **1**

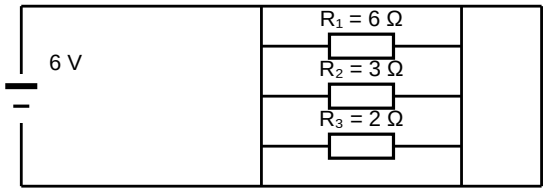


A. real, inverted and magnified, at 30 cm from the mirror
B. virtual, erect and magnified, behind the mirror
C. real, inverted and diminished, at 7.5 cm
D. real, inverted and of the same size, at 20 cm

31. When we move from bright sunlight into a dimly lit room we cannot see anything clearly at first, but after some time we start seeing. This is because: **1**

A. the pupil dilates and allows more light to enter the eye
B. the pupil contracts and allows less light to enter
C. the eye lens becomes thinner
D. the retina moves closer to the lens

32. Three resistors of 6 Ω, 3 Ω and 2 Ω are connected in parallel to a 6 V battery as shown. The total current drawn from the battery is: **1**



A. 1 A B. 3 A C. 6 A D. 11 A

33. **Assertion (A):** Tungsten is used for making the filament of an electric bulb. **1**
Reason (R): Tungsten has a very high melting point.

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, but R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

34. A lens has a focal length of −50 cm. Find its power and state its nature. **2**

OR

Why is copper used for connecting wires, while an alloy like nichrome is used in heating devices?

35. Explain how a rainbow is formed in the sky. Under what conditions is it seen? **3**

36. An electric iron of power 1.1 kW is operated on a 220 V supply for 2 hours every day. Calculate (a) the current drawn, (b) the energy consumed in 30 days and (c) the monthly bill at ₹6 per unit. **3**

37. What is meant by the magnetic field? State three properties of magnetic field lines. **3**

38. A dynamo fixed on the wheel of Gurpreet's bicycle lights the lamp when the cycle moves. Inside the dynamo a small permanent magnet rotates near a coil of wire. He notices that the lamp glows brighter when he cycles faster and stops glowing when he stops.

(a) Name the principle on which the dynamo works. 1

(b) What is the nature of the current produced by this dynamo, AC or DC? Justify. 1

(c) Why does the lamp glow brighter when the cycle moves faster? 2

OR

(d) State Fleming's right-hand rule and name the three fingers used.

39. A convex lens of focal length 12 cm forms an image of a 4 cm tall object placed 8 cm from the lens. Find the position, size and nature of the image and draw the ray diagram. Name one use of a lens in such a situation. **5**

OR

(a) Describe Faraday's experiment that demonstrates electromagnetic induction using a coil and a bar magnet. (b) State the rule used to find the direction of the induced current. (c) Write two differences between AC and DC.